

Racial Disparities in Crime Victimization in Los Angeles

Wenxin Cui

1. Introduction

The media often portrays crimes through individual narratives, making them appear as isolated events driven primarily by personal factors. However, examining crime through a broader social lens reveals systematic patterns in victimization across communities. Studies have documented substantial racial disparities in exposure to property and violent crimes in major U.S. cities, with Black and Latinx communities experiencing higher crime rates than White and Asian communities (Alba et al., 1994; Lodge et al., 2021). These disparities cannot be fully explained by individual socioeconomic factors, such as household income, suggesting that broader social and structural conditions shape unequal experiences of crime (Alba et al., 1994).

In light of these findings, this study analyzed crime records from the Los Angeles Police Department to investigate differences in crime victimization patterns across racial groups, with a particular focus on the distribution of crime types.

2. Data

2.1 Data Source

The data used in this study were obtained from the Los Angeles Police Department through the City of Los Angeles Open Data Portal. The dataset contains 1,004,894 police-reported crime incidents recorded in Los Angeles between January 2020 and December 2024, transcribed from original paper crime reports.

Each record corresponds to a single reported crime incident and contains 28 variables describing characteristics such as occurrence date, crime classification, victim demographics, geographic location, and case status.

2.2 Data Cleaning

Several preprocessing steps were performed to prepare the data for analysis. Irrelevant variables were removed, and records corresponding to attempted crimes, containing incomplete geographic information (i.e., latitude or longitude equal to zero), or reporting invalid victim ages (i.e., negative values) were excluded. The occurrence date of each incident was separated into year and month variables to facilitate temporal analyses, and categorical variables were converted to factors for statistical modeling.

The original dataset records victim race in 21 categories, which were consolidated into four broader racial groups for analysis. Likewise, the 140 crime classifications were grouped into five broader categories to reduce sparsity and improve the reliability of statistical comparisons.

After data cleaning, the dataset contained 660,130 crime incidents and 14 variables. The inferential analysis focused on victim race and crime type, while additional variables, including the date, time, and location of occurrence, were used to explore temporal and spatial patterns in victimization.

2.3 Exploratory Analysis

Racial differences in crime victimization were first explored visually. As shown in Figure 1, the distribution of crime types varied across racial groups. Property crimes accounted for a similar share of victimization across all groups. In contrast, sexual commodification offenses represented a relatively larger share among Hispanic victims, whereas violent crimes represented a relatively larger share among Asian victims.

	Commodified Sex	Financial	Property	Violent	Other
Asian	0.0034	0.096	0.20	0.45	0.25
Black	0.0044	0.11	0.22	0.34	0.33
Hispanic	0.0048	0.075	0.22	0.30	0.40
White	0.0045	0.094	0.23	0.38	0.29

Figure 1: Distribution of crime types among victims within each racial group.

Hispanic victims accounted for over 40% of victims at all hours of the day, whereas Asian victims consistently accounted for less than 5%. Across years, Hispanic victims continued to represent the largest share of victims among all racial groups, although this proportion declined notably in 2024. In contrast, the proportion of Asian victims increased during the same period.

A closer examination of the hourly composition of victims reveals distinct patterns across racial groups. Hispanic and White victims exhibited contrasting trends throughout the day: the share of Hispanic victims declined around noon, whereas the share of White victims peaked during the same period. Although both Hispanic and Black victims represented larger shares of victims during the early morning and late evening hours, Hispanic victims showed a more pronounced midday decline, while Black victims exhibited relatively less variation throughout the day.

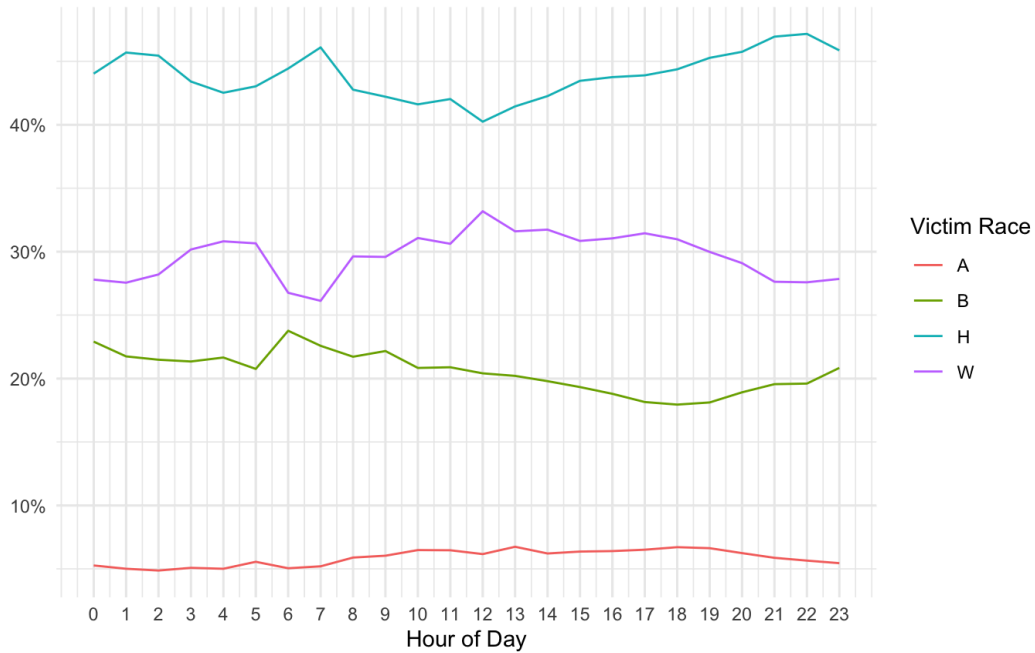


Figure 2: Racial composition of victims throughout the day.

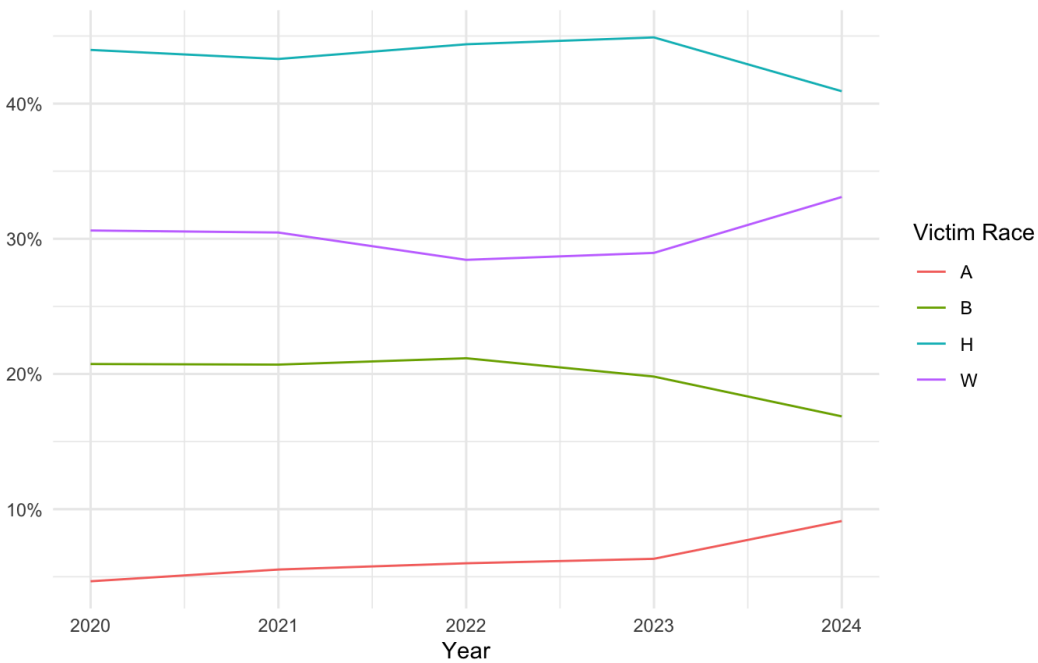


Figure 3: Racial composition of victims from 2020 to 2024.

Examining the geographical distribution of two primary crime types—violent and property crimes—reveals additional differences in victimization patterns across racial groups. Hispanic and White victims were more broadly distributed across central and northwest regions, with White victims also showing a notable presence along coastal areas, whereas Black and Asian

victims were more concentrated in the central region. Among property crime incidents, Hispanic victims represented a larger share of less severe offenses, while Asian victims had greater representation among more severe offenses.

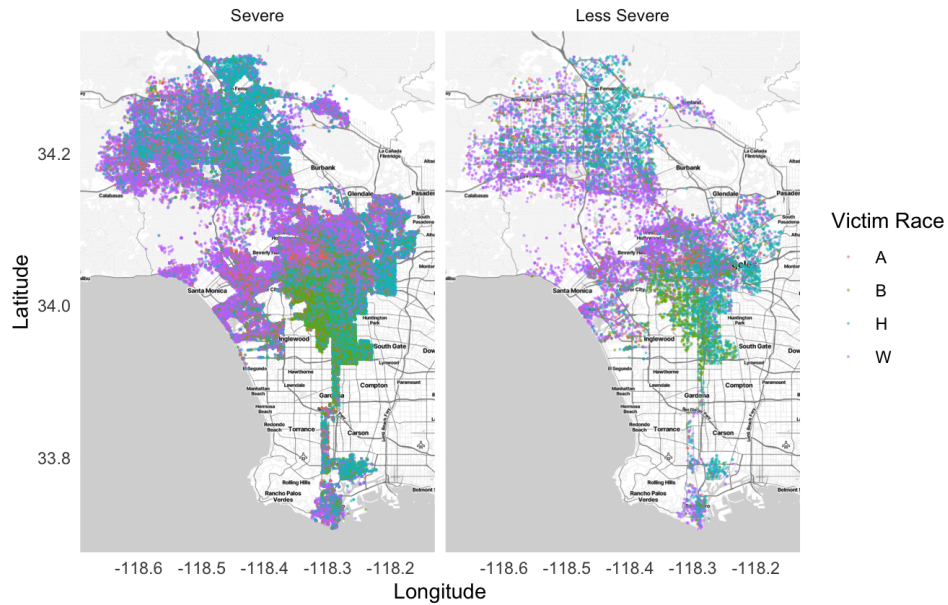


Figure 4: Geographical distribution of violent crimes by victim race and severity. Basemap tiles provided by Stadia Maps, Stamen Design, OpenMapTiles, and OpenStreetMap contributors.

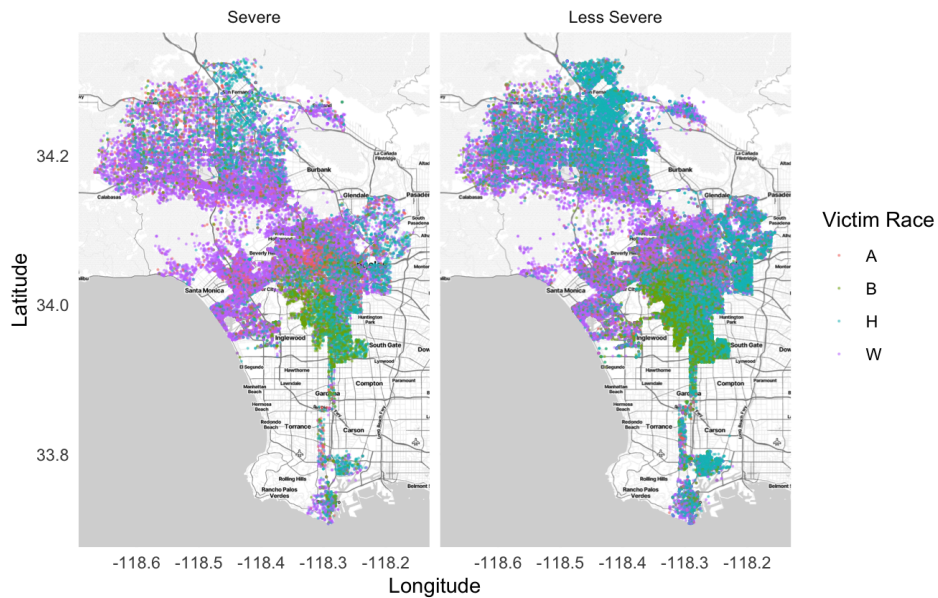


Figure 5: Geographical distribution of property crimes by victim race and severity. Basemap tiles provided by Stadia Maps, Stamen Design, OpenMapTiles, and OpenStreetMap contributors.

3. Methods

We conducted a series of two-sample permutation tests to assess the association between race and crime victimization.

3.1 Testing Framework

A two-sample permutation test is a non-parametric method used to evaluate whether two groups differ significantly. It tests the null hypothesis that observations are exchangeable between groups, meaning that group membership does not affect the distribution of the observations. The test generates a null distribution by randomly shuffling group labels and calculating a predetermined test statistic for each permutation. The observed test statistic is then compared with this null distribution to determine how likely the observed difference would be under the null hypothesis.

Here, we considered two categorical variables and applied a modified approach. We conducted two-sample permutation tests for all possible pairs of racial groups to assess whether the distribution of crime types differed between each pair.

Let P_i denote the probability distribution of crime types for racial group i , and let P_j be defined similarly for racial group j . For each pair of racial groups i and j , where $i \neq j$, we formulated the hypotheses as follows:

$$H_0 : P_i = P_j$$

$$H_a : P_i \neq P_j$$

We had four racial groups (i.e., White, Asian, Hispanic, and Black), resulting in $\binom{4}{2} = 6$ pairwise permutation tests. For each pairwise test, we held the crime types fixed and randomly shuffled racial group labels among victims, generating a null distribution of test statistics under the hypothesis that crime victimization was independent of race. The observed test statistic was then compared with this null distribution to assess the significance of the observed difference.

We adjusted p-values using the Bonferroni correction to account for multiple comparisons and control the family-wise error rate. The adjusted p-value was calculated as $\min(kp_{ij}, 1)$, where $k = 6$ was the total number of pairwise comparisons and p_{ij} was the unadjusted p-value from the permutation test for racial groups i and j . Statistical significance was determined by comparing the adjusted p-values with the chosen significance level.

3.2 Test Statistic

To measure the difference between two categorical distributions, we selected the total variation distance as the test statistic, which is defined as follows:

$$\text{TVD}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|$$

Total variation distance (TVD) quantifies the difference between two probability distributions. $P(x)$ and $Q(x)$ denote the probability mass functions of the discrete distributions P and Q at a given value x , with the sum taken over all possible values in the sample space. In our analysis, P and Q represent the crime type distributions for a given pair of racial groups, and x corresponds to a specific crime type. The test statistic ranges from 0 to 1, where a value of 0 indicates identical distributions and a value of 1 indicates completely disjoint distributions.

3.3 Rationale

Permutation tests offer a flexible approach for hypothesis testing without requiring large sample sizes or strong distributional assumptions. Although our dataset contained 660,130 crime incidents, comparisons across racial groups involved smaller and imbalanced samples.

Permutation tests are particularly useful in such settings, where asymptotic approximations may be unreliable. Unlike parametric methods, they do not require a specific underlying distribution and instead rely on the exchangeability of observations under the null hypothesis, making them well suited for our analysis.

Additionally, permutation tests allow flexibility in the choice of test statistic, enabling researchers to select measures that align with specific research questions. In our analysis, total variation distance was selected as the test statistic because it provided an intuitive measure of differences between crime type distributions.

Permutation tests are flexible but rely on several key assumptions. Under the null hypothesis, observations must be exchangeable, such that randomly reassigning group labels does not alter the distribution of the observations. They also require observations to be independent. Our exploratory analysis did not reveal systematic differences in the spatial distributions of observations across racial groups that would undermine the exchangeability assumption, and individual crime incidents were unlikely to exhibit meaningful dependence. Therefore, both assumptions were considered reasonable for this dataset.

4. Simulation Studies

We conducted simulation studies based on the original crime data to evaluate the operating characteristics of the pairwise permutation tests, focusing on type I error rates, statistical power, and robustness to violations of the exchangeability assumption.

4.1 Type I Error Rate

A type I error occurs when the null hypothesis is rejected despite being true. In this context, it corresponds to concluding that the crime type distributions differ between two racial groups when they are in fact identical.

To evaluate type I error rates, we simulated datasets of 100 observations in which crime type and race were independent. Specifically, racial group labels were generated first, after which crime types were independently sampled from a common probability distribution regardless of race. This simulation was repeated 1,000 times, and the type I error rate for each pairwise test was estimated as the proportion of simulations in which the null hypothesis was rejected.

The estimated type I error rates ranged from 0.09 to 0.10 across all pairwise tests. These values exceeded the nominal significance level of 0.05, suggesting that the tests were more likely than expected to identify differences in crime type distributions when none existed.

A vs B	A vs H	A vs W	B vs H	B vs W	H vs W
0.099	0.10	0.093	0.092	0.091	0.092

Figure 6: Estimated type I error rates of the pairwise permutation tests based on simulated data.

4.2 Statistical Power

Statistical power is the probability of correctly rejecting the null hypothesis when it is false. In this context, it corresponds to the probability of detecting differences in crime type distributions between two racial groups when such differences truly exist.

To evaluate statistical power, we simulated datasets of 100 observations in which crime type and race were associated. Specifically, a distinct probability distribution of crime types was specified for each racial group, and crime types were generated according to each victim's race. This simulation was repeated 1,000 times, and the statistical power of each pairwise test was estimated as the proportion of simulations in which the null hypothesis was rejected.

Statistical power varied across the pairwise tests. Comparisons involving Black victims exhibited high power, whereas the remaining comparisons showed low to moderate power, suggesting that the tests were less sensitive to true differences in crime type distributions among Asian, Hispanic, and White victims.

A vs B	A vs H	A vs W	B vs H	B vs W	H vs W
0.72	0.26	0.25	0.69	0.76	0.49

Figure 7: Estimated statistical power of the pairwise permutation tests based on simulated data.

4.3 Robustness to Violations of Exchangeability

We conducted additional simulation studies to evaluate the type I error rates of the pairwise tests when the exchangeability assumption was violated.

We simulated datasets of 100 observations in which race and crime type remained independent, but observations exhibited dependence. Specifically, dependence was introduced by assigning a higher probability to the crime type of the previous incident when generating each subsequent incident, creating sequentially correlated observations while preserving the overall independence between race and crime type. This simulation was repeated 1,000 times, and type I error rates were estimated as the proportion of simulations in which the null hypothesis was rejected.

The resulting type I error rates did not differ substantially from those obtained when exchangeability held and even decreased for tests comparing White victims with Black and Hispanic victims. This may suggest that either the test statistic was insensitive to the dependence structure introduced or that the induced dependence was too weak to affect the permutation tests.

A vs B	A vs H	A vs W	B vs H	B vs W	H vs W
0.095	0.11	0.093	0.092	0.081	0.088

Figure 8: Estimated type I error rates of the pairwise permutation tests under violations of exchangeability.

5. Results

The observed total variation distances were above 0 for all racial group comparisons, reflecting differences in crime type distributions across groups. However, the relatively small magnitudes (0.05 to 0.17) suggest that these differences were modest, with substantial overlap in victimization patterns. The variation in distances further indicates that some racial group comparisons showed greater divergence in crime type distributions than others.

A vs B	A vs H	A vs W	B vs H	B vs W	H vs W
0.11	0.17	0.069	0.075	0.050	0.10

Figure 9: Observed total variation distances for pairwise racial group comparisons.

For all racial group comparisons, the observed total variation distances fell in the extreme tails of their corresponding null distributions, with all adjusted p-values being virtually 0. These results indicate that the observed differences would be highly unlikely if crime type distributions were

unrelated to racial group membership, providing strong evidence that victimization patterns differed across racial groups. However, given the inflated type I error rates observed under simulated conditions, the statistical significance of these comparisons may be overstated.

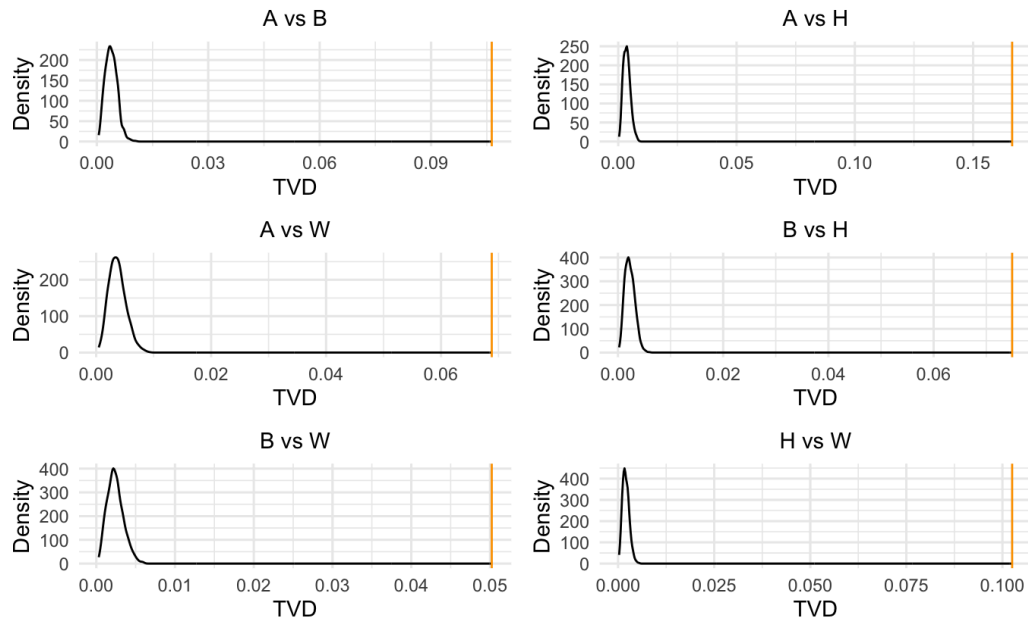


Figure 10: Observed total variation distances compared with their corresponding null distributions for each pairwise test. Null distributions are shown in black, and observed statistics are shown in orange.

6. Conclusion

Building on prior research documenting systematic racial disparities in victimization, this paper applied pairwise permutation tests based on total variation distance to compare crime type distributions among racial groups in Los Angeles.

The results provide strong evidence that the composition of crimes experienced differed by race, although the observed differences were modest in magnitude and the statistical significance may be overstated due to inflated type I error rates.

Bibliography

Adhikari, A., & DeNero, J. (2018). Two categorical distributions. *Computational and inferential thinking: The foundations of data science*. Cornell University.

<https://www.cs.cornell.edu/courses/cs1380/2018sp/textbook/chapters/16/1/two-categorical-distributions.html>

Alba, R. D., Logan, J. R., & Bellar, P. E. (1994). Living with crime: The implications of racial/ethnic differences in suburban location. *Social Forces*, 73(2), 395–434.

<https://doi.org/10.1093/sf/73.2.395>

Kahle, D., & Wickham, H. (2013). ggmap: Spatial visualization with ggplot2. *The R Journal*, 5(1), 144–161. <http://journal.r-project.org/archive/2013-1/kahle-wickham.pdf>

Lodge, E. K., Hoyo, C., Gutierrez, C. M., Rappazzo, K. M., Emch, M. E., & Martin, C. L. (2021). Estimating exposure to neighborhood crime by race and ethnicity for public health research. *BMC Public Health*, 21(1), Article 1078. <https://doi.org/10.1186/s12889-021-11057-4>

Los Angeles Police Department. (2026). Crime data from 2020 to 2024. *City of Los Angeles Open Data Portal*.

<https://data.lacity.org/Public-Safety/Crime-Data-from-2020-to-2024/2nrs-mtv8>